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| | **GLOBAL ECONOMICS**<br>**Grade: 10 - Learning Evidence 3**<br>**DECISION-MAKING UNDER UNCERTAINTY-C**<br>**Teacher: Nicolás López Cuéllar** | **1st TERM**<br>**2026–2027** |

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|---|---|
| **Learning objective:**<br>Compare decision criteria and their limitations, and construct and interpret decision trees for staged filtering under uncertainty. | **Assessment criteria:**<br>**C8**: Compares the Maximax, Maximin, Minimax Regret, and EMV/Bayes decision criteria by contrasting their information requirements, risk orientations, recommended alternatives, and limitations.<br>**C9**: Constructs and interprets decision trees for staged filtering rules, applying decision criteria in the specified order to identify the final alternative.<br>**C10**: Evaluates economic and financial decision-making scenarios by selecting and justifying the most appropriate decision criterion according to the information available, risk attitude, and decision context. |

**Name:** _____ **Grade:** _____ **Date:** _____

## Criteria assessment

This exam assesses C8 and C9, each worth 6 points. A **minimum of 4 points per criterion** is required to pass. Show all relevant calculations and justify your answers. Partial credit may be awarded for correct procedures despite arithmetic errors; answers without demonstrated procedures receive no credit. A 1-point penalty per criterion applies if the student's name is missing or the assessment is submitted late. C10 is assessed separately.

| № | C8 |
|---|---|
| 1 | 3 |
| 2 | 3 |
| **Total** | **6** |

| № | C9 |
|---|---|
| 3 | 2 |
| 4 | 2 |
| 5 | 2 |
| **Total** | **6** |

## 1. Camera Rental Plans [C8-3]

A camera rental studio selects its inventory under uncertain demand. Profits are in thousand USD.

| Alternative | High | Normal | Low |
|---|---|---|---|
| $P(S_j)$ | 0.2 | 0.5 | 0.3 |
| Essential ($A_1$) | 115 | 100 | 75 |
| Shared ($A_2$) | 155 | 165 | 45 |
| Extensive ($A_3$) | 265 | 55 | -30 |
| Modular ($A_4$) | 190 | 145 | -10 |

Decision statistics (Regret = maximum regret):

| Alt. | Max | Min | Regret | EMV |
|---|---|---|---|---|
| $A_1$ | 115 | 75 | 150 | 95.5 |
| $A_2$ | 165 | 45 | 110 | 127 |
| $A_3$ | 265 | -30 | 110 | 71.5 |
| $A_4$ | 190 | -10 | 85 | 107.5 |

Using the original camera rental payoffs, identify the alternative recommended by Maximax, Maximin, Minimax Regret, and EMV/Bayes, and report each criterion's deciding statistic value. **[1]** A subsidy $s$ (integer, $0 \le s \le 100$) increases the low-demand payoffs of $A_3$ and $A_4$ by $s$. An alternative is eligible only if all its payoffs are nonnegative. Determine each camera rental plan's eligibility based on the value of $s$. **[0.5]** Among eligible camera rental plans, find the smallest allowed subsidy $s$ at which the Maximin recommendation changes from $A_1$ to $A_4$ alone. Identify any tie at the threshold and distinguish it from a strict change. Respect the integer domain $0 \le s \le 100$; if no smallest value exists, state the threshold and the condition for a change. **[1.5]**

## 2. Outdoor Cinema Refreshments [C8-3]

A cinema manager selects a refreshment kiosk under uncertain audience size. Profits are in hundred USD.

| Alternative | High | Low |
|---|---|---|
| $P(S_j)$ | 0.3 | 0.7 |
| Portable ($C_1$) | 108 | 72 |
| Full-service ($C_2$) | 174 | 22 |
| Deluxe ($C_3$) | 204 | -8 |
| Fixed ($C_4$) | 158 | 54 |

Decision statistics (Regret = maximum regret):

| Alternative | Max | Min | Regret | EMV |
|---|---|---|---|---|
| $C_1$ | 108 | 72 | 96 | 82.8 |
| $C_2$ | 174 | 22 | 50 | 67.6 |
| $C_3$ | 204 | -8 | 80 | 55.6 |
| $C_4$ | 158 | 54 | 46 | 85.2 |

Using the original refreshment kiosk payoffs, identify the alternative recommended by Maximax, Maximin, Minimax Regret, and EMV/Bayes, and report each criterion's deciding statistic value. **[1]** A sponsor replaces $C_1$'s large-audience payoff of 108 with $g$ (integer, $108 \leq g \leq 180$). An alternative is eligible only if all its payoffs are at least 46. Determine each refreshment kiosk's eligibility based on the value of $g$. **[0.5]** Among eligible refreshment kiosks, find the smallest allowed sponsored payoff $g$ at which the Minimax Regret recommendation changes from $C_4$ to $C_1$ alone. Recalculate regret benchmarks using only eligible kiosks. Identify any tie at the threshold and distinguish it from a strict change. Respect the integer domain $108 \leq g \leq 180$; if no smallest value exists, state the threshold and the condition for a change. **[1.5]**

## 3. Garden Centre Stock [C9-2]

A garden centre selects a seedling stock plan $D_1$–$D_6$. Net profits are in hundred USD. The decision statistics are summarized below:

| Alternative | Max | Min | Regret | EMV |
|---|---|---|---|---|
| $D_1$ | 164 | 68 | 72 | 137.6 |
| $D_2$ | 188 | 44 | 48 | 154.4 |
| $D_3$ | 140 | 92 | 96 | 123.2 |
| $D_4$ | 212 | 20 | 72 | 154.4 |
| $D_5$ | 116 | 68 | 120 | 99.2 |
| $D_6$ | 236 | -4 | 96 | 154.4 |

Apply these stages in order: EMV/Bayes: Best 3. Maximax: Best 2. Maximin: Final selection. Illustrate each filtering stage, showing survivors and eliminations.

## 4. Climbing Gym Expansion [C9-2]

A climbing gym selects an expansion plan $F_1$–$F_6$. Net profits are in thousand USD. The decision statistics are summarized below:

| Alternative | Max | Min | Regret | EMV |
|---|---|---|---|---|
| $F_1$ | 213 | 59 | 55 | 154.7 |
| $F_2$ | 169 | 103 | 99 | 144.8 |
| $F_3$ | 191 | 81 | 77 | 164.6 |
| $F_4$ | 268 | 4 | 99 | 138.2 |
| $F_5$ | 147 | 92 | 121 | 125 |
| $F_6$ | 235 | 37 | 66 | 145.9 |

Apply these stages in order: Minimax Regret: Best 5. EMV/Bayes: Best 4. Maximax: Best 2. Maximin: Final selection. Illustrate each filtering stage, showing survivors and eliminations.

## 5. Art Materials Orders [C9-2]

An art materials distributor selects an order plan $H_1$–$H_6$. Net profits are in hundred USD. The decision statistics are summarized below:

| Alternative | Max | Min | Regret | EMV |
|---|---|---|---|---|
| $H_1$ | 129 | 81 | 72 | 111.4 |
| $H_2$ | 145 | 65 | 56 | 124.2 |
| $H_3$ | 177 | 33 | 48 | 112.2 |
| $H_4$ | 113 | 73 | 88 | 97 |
| $H_5$ | 201 | 9 | 72 | 106.6 |
| $H_6$ | 161 | 49 | 40 | 118.6 |

Apply these stages in order: Maximin: Best 4. Minimax Regret: Best 3. Maximax: Best 2. EMV/Bayes: Final selection. Illustrate each filtering stage, showing survivors and eliminations.

## C10: Separate activity reminder (unscored)

Complete the individual **Interactive Decision Webpage: C10 Practice** activity and submit the deployed webpage URL. See the separate activity for instructions.